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Unit – 1 : Session – 6 : SLO – 1 | Process Life Cycle — Fill in the Missing Words

■ I. Completed Sentences (Process Life Cycle)

- A PROCESS is a program in execution.
- A process passes through states: NEW, READY, RUNNING, WAITING (BLOCKED) and TERMINATED.
- The OS creates a process and assigns it a PROCESS CONTROL BLOCK (PCB) holding its state, program counter, registers and memory information.
- The SCHEDULER (dispatcher) picks a ready process and gives it the CPU.
- When a running process requests I/O it moves to the WAITING state.
- After I/O completes, the process returns to the READY queue.
- When a process finishes execution it is TERMINATED and its resources are deallocated.
- CONTEXT SWITCHING saves the state of one process and restores the state of another.

■ II. Role of the OS

- Creation: fork()/spawn, PCB allocation, memory setup.
- Scheduling: choosing which ready process runs next (short-term scheduler).
- Termination: exit(), releasing memory, closing open files, removing the PCB.